HUMSAFAR

Hardware Selection & Integration Document

Prepared for supervisor review prior to hardware procurement — Implementation Preparation Phase
Based on the locked Humsafar SRS v1.0 and SDS v1.0

Executive Summary

Humsafar's locked requirements call for exactly two physical hardware integrations: a wearable health band (FR-H2, supervisor-mandated) and a camera for on-demand viewing (FR-K1, FR-K2). The team has **officially decided** to use a spare Android phone as the camera source rather than purchasing a dedicated IP camera, and this document now reflects that as the locked implementation approach. The wearable band has **not yet been finalized** — the team is comparing wristbands, armbands, and chest straps in local markets before purchasing; this document retains reference candidates and purchasing criteria until a specific model is chosen.

Document status: Section 1 (Camera) is LOCKED. Section 2 (Wearable Band) remains OPEN, pending final device selection.

1. Wearable Health Band — STATUS: OPEN (Pending Final Model Selection)

The team has decided to visit local markets to compare wristbands, armbands, and chest straps in person before purchasing, rather than committing to a specific model from research alone. This section will be finalized once a device is selected. The criteria and reference candidates below guide that decision; they are not a final recommendation.

Official Purchasing Criteria

- Support for standard Bluetooth Low Energy (BLE)
- Standard GATT Heart Rate Service (0x180D)
- Easy Android integration
- Affordable price
- Reliable availability
- Comfortable enough for demonstration

Reference Candidates (for market comparison — not a final pick)

Type	Example	Price (approx.)	Note
Chest strap	CooSpo H6 / Magene H64	$16-30	Cheapest, simplest integration; less comfortable for continuous/all-
Armband (optical)	CooSpo HW9	$40-60	Best comfort-to-cost balance for a Dependent who wears it all day,
Wrist-worn	Amazfit (Zepp OS 3.0+ models)	$40-90	Verify in-store: some models only broadcast HR during an active w
Arm/temple	Polar Verity Sense	$90-100	Premium reliability, continuous broadcast, higher cost.

Verification step (mandatory before purchase, any model): use a free BLE scanner app (e.g., nRF Connect for Mobile) to confirm the device exposes Heart Rate Service (0x180D) and Battery Service (0x180F) before finalizing the purchase.

Once a model is purchased, this document will be updated with: exact specifications, final cost, and a confirmed Android integration plan (BLE library, background sync design) tailored to that specific device.

2. On-Demand Camera (for Live View and Event Clips) — STATUS: LOCKED

Official decision: the team will use a spare Android phone as the camera device. A dedicated IP/CCTV camera will not be purchased. This is the locked implementation approach and will only be revisited if the supervisor explicitly requests a standalone IoT camera device.

Why Required	Enables the Guardian's on-demand live view and event-triggered clip capture of the Dependent's environment.
Feature(s) Supported	FR-K1 (On-Demand Live Camera View), FR-K2 (Event-Triggered Clip Capture), FR-N1 (end-to-end encryption of the stream).
Buy vs. Build	**BUILD (software-only) — LOCKED.** Official decision: use a spare Android phone as the camera source. No dedicated IP camera will be purchased. *Suggested refinement given the Flutter stack:* rather than relying on a third-party IP-camera app, consider building a lightweight "camera mode" screen directly in your own Flutter app (using the camera plugin plus a simple streaming approach) so the whole system stays in one codebase — this is also better academic value, since you're building the streaming capability rather than using someone else's app.
Recommended Specifications	Android 8+, working rear camera, reliable Wi-Fi, kept powered and mounted during testing/demo.
Estimated Cost	**PKR 0** (reuse an existing device). No dedicated camera purchase planned.
Communication Protocol	Wi-Fi (local network); HTTP or RTSP video stream from the spare phone.
Android Integration	Guardian app requests a stream reference from the backend (**GET /dependents/{id}/camera/stream**) and renders it using a Flutter video-player package. If the team builds its own camera-mode screen (see Buy vs. Build), the spare phone runs the same Flutter codebase in a restricted "camera" mode.
App-to-Backend Communication	The backend's CameraStreamService brokers access permission and, where applicable, proxies or references the stream; captured event clips upload to object storage via ClipCaptureService (per SDS).
Implementation Complexity	**Medium.** Manageable within the team's Flutter skillset; no native RTSP/ONVIF protocol work required.
Academic Value	**Medium-High.** Demonstrates real-time media streaming and permission-gated access control, built entirely by the team rather than relying on a third-party app.
Risks & Practical Considerations	Camera and Guardian device should be on the same local network for a straightforward demo, or a relay/TURN-style approach is needed for remote access. The on-screen "being viewed" indicator (FR-K1.3) must be implemented regardless of streaming method.
Better Alternatives	None planned. A dedicated ONVIF-compliant Wi-Fi camera remains available only if the supervisor specifically requests a standalone IoT hardware demonstration.
Final Recommendation	**Locked:** spare Android phone as the camera source, for both development and the FYP demo.

Note on hardware NOT required: Panic Mode (FR-D2), Remote Alarm Override (FR-D3), Fall Detection (FR-D6), and location tracking (FR-C1) all use sensors and actuators already built into a standard Android phone (touchscreen, speaker, accelerometer, GPS). No additional hardware is needed for these features, and none is recommended.

3. Supporting Infrastructure (Not a Purchase Item)

Item	Required?	Est. Cost	Notes
Android test devices (2-3)	Yes	PKR 0 (team-owned)	Android 8.0+ (API 26), BLE 4.0+, working GPS/camera/mic. Emulators cann
Wi-Fi network	Yes	PKR 0 (existing)	Needed for camera streaming and general connectivity during development a
Backend hosting	Yes	Free tier likely sufficient	Cloud-hosted per SDS Section 16; no dedicated server hardware purchase r

4. Overall Hardware Architecture

The Dependent's Android phone is the hub: it connects to the wearable band over BLE and (in the spare-phone approach) is itself the camera source or connects to one over local Wi-Fi. The phone communicates with the cloud-hosted backend over HTTPS, which in turn pushes data and stream access to the Guardian's Android phone. No hardware communicates directly with the backend — the Dependent phone is always the intermediary, consistent with the SDS's client-server architecture.

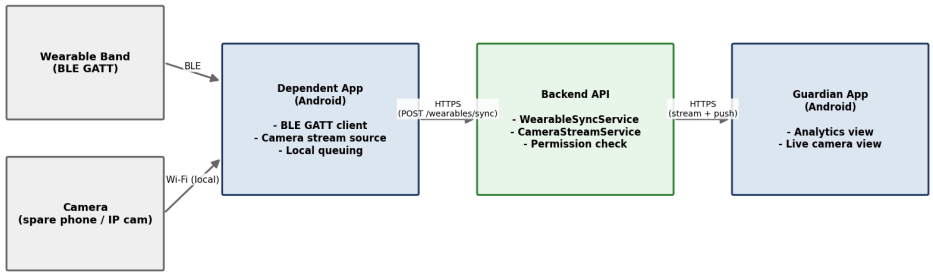
5. Hardware-to-Software Integration Flow

1. Wearable band pairs with the Dependent phone over BLE (standard GATT bonding).
2. Dependent app reads Heart Rate and Battery characteristics periodically via a background sync service.

3. Dependent app sends synced readings to the backend (POST /wearables/sync) over HTTPS/TLS.
4. Camera (spare phone or dedicated device) streams over the local Wi-Fi network.
5. Guardian app requests stream access from the backend, which checks permission before authorizing.
6. Backend returns a stream reference; the Guardian app renders the live video.
7. Backend stores wearable data and clip metadata, and drives analytics and notifications as configured.

6. Data Flow: Sensor / Camera to Guardian

Data Flow: Sensor / Camera → Mobile App → Backend → Guardian



7. Hardware Procurement Checklist

#	Item	Action Before Purchase	Est. Cost
1	BLE heart-rate band	Confirm open GATT access (Heart Rate + Battery services) from listing/seller	PKR 3,000-7,000
2	Spare Android phone (camera)	Confirm a team member has a spare device; install an IP-camera app	PKR 0
3	Android test devices (x2-3)	Confirm Android 8.0+, BLE 4.0+ on team-owned phones	PKR 0
4	Dedicated IP camera	Not planned. Only revisit if supervisor explicitly requests a standalone IP camera device	PKR 0

8. Priority Order for Purchasing

1. **Confirm test devices first** (no cost, no lead time) — verify at least 2-3 team-owned Android phones meet the minimum spec.
2. **Set up the spare phone as the camera source** — locked decision, no purchase needed, can start immediately.
3. **Visit local markets to compare wearable bands** against the official purchasing criteria (Section 1) before buying — the only remaining open hardware decision.

9. Estimated Total Budget

Scenario	Estimated Cost
Locked plan (spare phone camera + wearable TBD)	PKR 4,500 - 9,000 (~US$16-30) — camera cost is PKR 0; wearable e
If supervisor requests a dedicated IP camera (not currently planned)	Add PKR 3,000 - 8,000

10. What Can Be Developed Immediately (No Hardware Needed)

Nearly the entire application: UI/UX for both apps, authentication and KYC flow (against a mocked verification response), the full backend API and database, the permission dashboard, medication scheduling and escalation logic, geofencing, notifications, medical records and analytics screens, and the messaging/calling UI. All of this can be built and tested using

emulators or plain Android devices, with wearable and camera data simulated/mocked.

11. What Must Wait Until Hardware Is Available

Real BLE pairing, GATT reads, and background-reconnection tuning (the actual FR-H2/NFR-13 behavior); true end-to-end camera streaming and the on-screen viewing-indicator behavior (FR-K1.3); and final battery/performance validation on physical devices (NFR-02). These require the physical band and at least two real Android phones, and should be scheduled only after the core application logic is stable.

12. Final Recommendations Before Purchasing

- Verify open GATT access (nRF Connect) on the specific wearable band before paying — this is the single most likely purchase mistake.
- Camera decision is locked: spare Android phone, no dedicated purchase. Only revisit if the supervisor explicitly requests a standalone IoT camera.
- Do not over-invest in camera hardware even if revisited: the locked scope already excludes continuous 24/7 recording (Future Scope), so an expensive, feature-rich camera would be spending against requirements that were deliberately trimmed.
- Inform the technical advisor of the exact wearable model once purchased, so BLE integration guidance can be tailored to that device's specific GATT implementation.
- No hardware purchase should block starting development; only BLE/camera-specific modules are hardware-dependent (see Sections 10-11).